# R Programming Examination

Coenn Johan Capalihan

2026-11-09

Part A: Basic Operations

A1

A2

A3

```
17 %% 5
```

```
## [1] 2
```

```
17 %/%5
```

```
## [1] 3
```

```
2^10
```

```
## [1] 1024
```

```
sqrt(144)
```

```
## [1] 12
```

```
round(pi, 0.001)
```

```
## [1] 3
```

A4

```
a <- c(2,4,6,8)
b <- c(1,3,5,7)

a+b
```

```
## [1]  3  7 11 15
```

```
a*b
```

```
## [1]  2 12 30 56
```

```
a>b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
sum(a*b)
```

```
## [1] 100
```

A5

```
paste("Student-1", "Student-2", "Student-3")
```

```
## [1] "Student-1 Student-2 Student-3"
```

```r
typeof(42)
```

```
## [1] "double"
```

```r
typeof(42L)
```

```
## [1] "integer"
```

```r
typeof("42")
```

```
## [1] "character"
```

```r
typeof(41i)
```

```
## [1] "complex"
```

```r
typeof(TRUE)
```

```
## [1] "logical"
```

Part B:Data Frames and Conditional Statements

B1

```r
students <- data.frame(
    id = c(1:8),
    name = c("Alice", "Bruno","Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
    sex = c("F","M","M","F","F","M","F", "M"),
    section = c("A","B","A","C","B","C","A", "B"),
    midterm = c( 78,85,92,67,74,88,95,59),
    final = c( 82,79,96,71,80,91,93,65),
    attendance = c(95,88,100,72,90,85,98,60),
    stringsAsFactors = FALSE
)
```

B2

```r
 str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```r
dim(students)
```

```
## [1] 8 7
```

```r
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```r
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
```

```
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
average <- round((students$midterm + students$final)/2)
average
```

```
## [1] 80 82 94 69 77 90 94 62
```

```
students<- cbind(students, average)
students
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95      80
## 2  2 Bruno   M       B      85    79         88      82
## 3  3  Chen   M       A      92    96        100      94
## 4  4  Dana   F       C      67    71         72      69
## 5  5  Elif   F       B      74    80         90      77
## 6  6 Farid   M       C      88    91         85      90
## 7  7 Grace   F       A      95    93         98      94
## 8  8  Hiro   M       B      59    65         60      62
```

B4